IITG

KRITI'25

Semi-Supervised Weed Detection Challenge

Organizer :
4i Labs



Low Prep
| 250 Pts

Semi-Supervised Weed Detection Challenge

250 Points

Start : 28/01

End : 05/02

Introduction:

Accurate and efficient weed detection is a cornerstone for automating agricultural tasks, such as targeted weed removal, leading to increased productivity and reduced manual labor. Training robust object detection models typically demands vast amounts of meticulously labeled data. Acquiring and annotating such datasets is an expensive and time-consuming bottleneck in developing practical AI solutions for agriculture.

We challenges you to overcome this limitation by exploring the power of semi-supervised learning. You will leverage a combination of a small labeled dataset and a larger pool of unlabeled data to train an effective weed detection model. This approach aims to significantly reduce the reliance on extensive labeled data while achieving high accuracy in identifying and localizing weeds within agricultural images.

Crucially, you are expected to demonstrate the effective utilization of the provided unlabeled data to achieve performance gains compared to a baseline model trained solely on the labeled data. The use of any external datasets is strictly prohibited.

Dataset Link

Problem Statement:

Your task is to develop a weed detection model using semi-supervised learning techniques. You are provided with:

1. A small labeled dataset of agricultural field images containing sesame crops and weeds (200 images).
2. A larger unlabeled dataset of similar images of sesame crops and weeds (1000 images).
3. A test dataset having 100 images and their corresponding annotations.

The objective is to train a model/s capable of accurately identifying and localizing weeds within these images, demonstrating a clear improvement in performance by effectively utilizing the unlabeled data.

Metric for evaluation: 0.5 * (F1-Score) + 0.5 * (mAP@[.5:.95])

Deliverables:

1. Well-commented and organized code: Include all necessary scripts for data preprocessing, training and inference on the test data using the provided metric.
2. Your trained model file(s).
3. A concise report (maximum 2 pages):
   - Clearly explain the semi-supervised learning technique(s) you implemented.
   - Describe your training methodology, any data augmentations used and your results.
   - Discuss any challenges you faced and how you addressed them.
   - Explain why you chose your specific approach and any insights gained
4. GitHub link of all the above deliverables should be submitted.

Judging Criteria:

1. Model Performance, Latency(inference time) - 150 points :
   metric: 0.5 * (F1-Score) + 0.5 * (mAP@[.5:.95]).
1. Code Submission Quality and Methodology - 50 Points
2. Report quality - 50 points
3. Submissions that demonstrate a significant performance improvement by utilizing the unlabeled data compared to a baseline model trained only on the labeled dataset will be judged more favorably. Provide clear evidence and comparisons in your report.

Rules:

1. The Participating team can have maximum of 4 members which should include at least one fresher.
2. The use of any external datasets is strictly prohibited.
3. Submission Deadline: 5th February 2025.
4. Submission Method: GitHub Repository Link. A dedicated submission form will be made available on the competition's end date